

Where and How Fast Is Heat Stress Worsening? A District-Level, Multi-Index Machine-Learning Projection for 64 districts of Bangladesh

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Abstract

Rising heat and humidity increasingly threaten health, labour, and energy systems across Bangladesh, yet operational planning needs district-resolved, uncertainty-aware projections of the specific conditions that drive human heat stress. We present a deterministic, reproducible machine-learning framework that projects seven temperature-correlated heat-stress conditions—heat-index comfort, wet-bulb temperature, indoor and outdoor wet-bulb globe temperature, cooling degree-days, ultraviolet exposure, and dew-point moisture stress—for all 64 districts of Bangladesh. Four gradient-boosting ensembles are trained on the 2014–2024 monthly record using future-known seasonal features; the observed 1980–2024 Theil–Sen trend is superimposed so that projections evolve realistically, and all thermal indices are recomputed from the projected primary variables to avoid target leakage. Validated against withheld 2025 observations, the projected indices attain median R^2 of 0.95–0.98, and their operational risk classes match observations

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to within one class in 100% of district-months. A Mann–Kendall analysis shows maximum temperature rising in every district (median $+0.56^{\circ}\text{C}$ per decade) while minimum temperature remains flat, indicating a widening diurnal range; the coastal and southwestern zones carry the greatest combined burden, and ultraviolet and oppressive-humidity hazards are effectively nationwide. District choropleths, zonal envelopes, and calibrated uncertainty bands provide a directly actionable heat-stress outlook while transparently bounding the models’ skill relative to climatology.

Keywords: heat stress, wet-bulb temperature, WBGT, seasonal projection, gradient boosting, Mann–Kendall trend, Bangladesh, climate adaptation

1. Introduction

Anthropogenic climate change has fundamentally altered the frequency, intensity, and duration of extreme heat events worldwide. Mean surface air temperatures have risen steadily since the pre-industrial era, and this warming has been accompanied by a disproportionate increase in heatwaves, prolonged humid spells, and compound heat–humidity extremes that the human body increasingly struggles to tolerate [1]. Unlike many climate hazards, extreme heat acts as a silent stressor, producing cumulative physiological strain that is frequently underreported in official mortality and morbidity statistics yet carries substantial public-health, occupational, and economic consequences [2]. Understanding how heat stress manifests across different climatic and geographic settings has therefore become an urgent priority for adaptation planning.

Air temperature alone does not represent the heat stress the human body actually experiences and can lead to misleading assessments. Relative humidity, solar radiation, wind, and atmospheric moisture all govern

17 the body’s ability to shed heat through evaporation, radiation, and convec-
18 tion; elevated humidity in particular suppresses the evaporation of sweat—
19 the body’s primary cooling mechanism—so that core temperature rises even
20 when the air is not exceptionally hot [3]. To capture these combined effects,
21 researchers have developed composite heat-stress indices that now underpin
22 public-health warning systems, occupational heat-safety standards [4], and
23 cooling-energy demand projections [5], and that increasingly recognise the
24 approach of physiological survivability limits under extreme humid heat [6].

25 South Asia is among the regions most acutely exposed to escalating heat
26 stress, owing to its dense population, high monsoon-season humidity, and
27 large share of outdoor and informal-sector labour. Bangladesh occupies a
28 uniquely exposed position within it: a tropical monsoon climate, low-lying
29 deltaic geography, and exceptionally high population density combine with
30 rapid, unplanned urbanization to let heat stress intensify quickly and dis-
31 tribute unevenly across the landscape [7]. Critically, the country is not
32 climatically homogeneous—its territory spans drought-prone northwestern
33 plains with continental microclimates, a densely built and heat-retentive cen-
34 tral urban core exhibiting strong urban heat-island effects, low-lying coastal
35 districts shaped by maritime humidity, seasonally inundated wetland basins,
36 and elevated hill tracts with distinct topographic regimes. Each zone is
37 expected to experience thermal stress through different pathways and at
38 different intensities, yet most existing heat-risk assessments in Bangladesh
39 continue to treat the country as a single climatic unit or rely on a narrow
40 selection of indices—most commonly ambient temperature alone—without
41 accounting for the multi-dimensional nature of heat exposure [8].

42 A parallel gap exists in temporal scope and forecasting capability. Most
43 prior work on heat stress in Bangladesh is retrospective in orientation or

44 resolved only to divisional or national summaries, and very few studies in-
 45 tegrate long-term climatological baselines with near-term machine-learning
 46 forecasting to project how thermal stress will evolve at the district level [**verify-**
 47 **intended**]. This omission matters, because heat-risk management—whether
 48 through early-warning systems, occupational health policy, or urban cooling
 49 infrastructure—requires knowing not only where risk is highest today but
 50 where and how quickly it is worsening. A recent wet-bulb globe tempera-
 51 ture analysis across Bangladesh reported warming of roughly $0.08\text{--}0.5^\circ\text{C}$ per
 52 decade, most pronounced in the southeast and northeast [9], underscoring
 53 the need for district-resolved, multi-index projection.

54 This study addresses these gaps through the Temperature-Related
 55 Conditions (TRC) framework: an integrated set of seven temperature-dependent
 56 indicators —heat-index comfort, wet-bulb temperature, indoor and outdoor
 57 wet-bulb globe temperature, heatstroke risk, cooling degree-days, ultraviolet
 58 exposure, and dew-point moisture stress—each capturing a distinct physio-
 59 logical, biophysical, or infrastructural dimension of heat-related risk. Apply-
 60 ing this framework to 45 years (1980–2024) of district-level meteorological
 61 records across all 64 districts and six climatically distinct zones, we (i) train
 62 an ensemble of gradient-boosting models [10] on future-known seasonal fea-
 63 tures and re-impose the observed long-term trend so that projections evolve
 64 realistically rather than collapsing onto the last training year; (ii) recom-
 65 pute every thermal index from the projected primary variables rather than
 66 predicting the indices directly, eliminating target leakage and preserving the
 67 physical coupling among temperature, humidity, and radiation; (iii) gener-
 68 ate 24-month district-level projections (2025–2026) with analytically prop-
 69 agated uncertainty bands; and (iv) validate the projections out-of-sample
 70 against independent 2025 observations, benchmark their skill transparently

71 against seasonal-naive and climatological baselines, and characterise long-
72 term change with the non-parametric Mann–Kendall test. The best model
73 per variable is selected by a multi-attribute simple-additive-weighting scheme
74 with a generalization-gap penalty [11].

75 Two features distinguish the analysis. First, results are reported at full
76 district resolution through choropleth maps rather than aggregated away,
77 exposing spatial gradients—such as the coastal concentration of humid-heat
78 burden—that zonal summaries obscure. Second, we are explicit about the
79 limits of predictive skill: at monthly resolution the ensembles essentially
80 match climatology, so their value lies in providing spatially complete, uncertainty-
81 aware, and trend-consistent projections of physiologically relevant indices
82 rather than in reducing monthly error. One indicator, heatstroke risk, was
83 found to lack out-of-sample skill and is retained only descriptively. The re-
84 mainder of this paper details the data and methods, presents the validated
85 projections and observed trends, and discusses their implications for heat-
86 adaptation planning across Bangladesh.

87 **2. Materials and Methods**

88 This study builds a deterministic, reproducible pipeline that (i) learns
89 the monthly climatology of seven primary meteorological variables for each
90 of the 64 districts of Bangladesh from a recent, high-quality window; (ii) su-
91 perimposes the long-term observed climate trend so that projections evolve
92 realistically beyond the training period; (iii) recomputes physiologically rel-
93 evant thermal indices from the projected primary variables to avoid target
94 leakage; (iv) quantifies projection uncertainty by first-order error propaga-
95 tion; and (v) validates the projected 2025 fields against independent obser-

96 vations. All methodological constants are fixed a priori and are reported
97 here so that the workflow can be audited and reproduced exactly.

98 2.1. Study area and climatic zonation

99 The analysis covers all 64 administrative districts of Bangladesh. For syn-
100 thesis and reporting, districts are grouped into six climatic zones: (1) North-
101 west Extreme Heat Zone (dry hot plains, including the Barind Tract), (2) Cen-
102 tral Urban Heat Zone (urban heat-island region), (3) Southwest Hot-Dry
103 Zone (drought- and salinity-affected areas), (4) Coastal Humid Heat Zone
104 (the coastal belt), (5) Haor/Wetland Humid Zone (the haor basin), and
105 (6) Hill Tract Zone (elevated terrain). The zonation is author-defined but
106 grounded in converging, established frameworks: the agro-climatic regionali-
107 sation of the Bangladesh Meteorological Department and the agro-ecological/physiographic
108 zonation of Bangladesh [12, 13], together with recognised hydro-ecological
109 units (Barind Tract, Sundarbans coastal belt, haor basin). Each district
110 is flagged as a *confirmed* (high-confidence) or *provisional* assignment, and
111 one representative *exemplar* district per zone is retained for main-text fig-
112 ures (Figure 1). The complete district-to-zone mapping is provided in the
113 supplementary material.

114 2.2. Data and preprocessing

115 Daily meteorological records were obtained from the Visual Crossing
116 Weather API [14] for each district. Seven primary (physically measured)
117 variables are used: mean, maximum, and minimum near-surface air temper-
118 ature ($^{\circ}\text{C}$); relative humidity (%); dew-point temperature ($^{\circ}\text{C}$); downward
119 solar (shortwave) radiation; and the ultraviolet (UV) index. The historical
120 record spans 1980–2024; an additional file covering 2025 (through 20 Novem-

Climatic zonation of Bangladesh (64 districts, 6 zones)

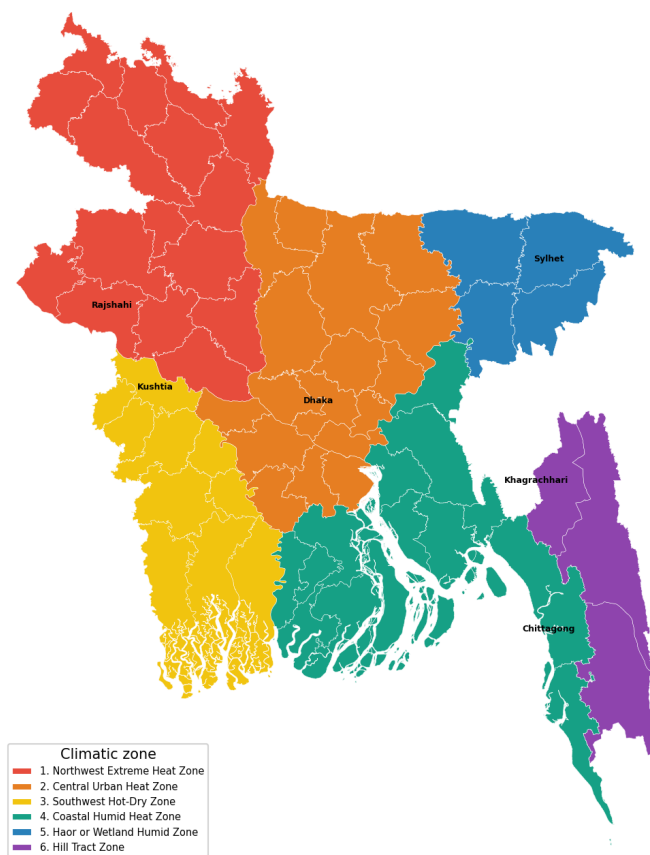


Figure 1: Climatic zonation of Bangladesh: the 64 districts grouped into six zones, with the exemplar district of each zone labelled.

ber 2025; the final month is partial) is reserved exclusively for out-of-sample validation.

Because humidity is noisy in the earliest decades and solar/UV are unavailable before 2014, the machine-learning window is restricted to 2014–2024, whereas the full 1980–2024 record is used only for the observational trend analysis (Section 2.12). Daily values are aggregated to calendar-month means (month-start resampling), and residual gaps are filled by linear interpolation. All modelling is performed on these monthly series.

2.3. Feature construction

To respect a strict *future-known-features-only* constraint—no lags, rolling windows, or autoregressive terms that would be unavailable at projection time—each month is described by a deterministic seasonal–secular feature vector. Seasonality is encoded by a truncated Fourier series with period $P = 12$ months and $H = 3$ harmonics,

$$\sin_h(m) = \sin\left(\frac{2\pi hm}{P}\right), \quad \cos_h(m) = \cos\left(\frac{2\pi hm}{P}\right), \quad h = 1, \dots, H, \quad (1)$$

where m is the calendar-month index. The secular component is the calendar year. The full design vector is therefore $\mathbf{x} = [\text{year}, \sin_1, \cos_1, \dots, \sin_3, \cos_3]$ (seven features).

2.4. Predictive models

Four gradient-based tree ensembles are trained independently for every district–variable pair: Random Forest [10], XGBoost [15], LightGBM [16], and CatBoost [17]. Hyperparameters are fixed a priori and applied uniformly across all districts and variables so that results are mutually comparable and the pipeline is fully deterministic (global random seed = 42). The fixed configurations are listed in the supplementary material.

145 2.5. Trend-aware detrending and projection

146 Tree ensembles cannot extrapolate beyond the range of their training fea-
 147 tures; if the calendar year is used directly as a predictor, every projected year
 148 collapses onto the last training year, producing a static (trendless) forecast.
 149 To carry the observed climate trend forward while preserving the future-
 150 known constraint, each primary variable is detrended before learning and
 151 re-trended at prediction time. For district-variable series $y(t)$ with continu-
 152 ous time t (in years), a Theil–Sen slope $\hat{\beta}$ is estimated from the annual-mean
 153 series,

$$\hat{\beta} = \text{median}_{i < j} \frac{\bar{y}_j - \bar{y}_i}{t_j - t_i}, \quad (2)$$

154 where $\bar{y}_k$ is the mean of year k . To keep the projected trend consistent with
 155 the manuscript’s observational trend analysis, $\hat{\beta}$ is taken from the robust
 156 1980–2024 record for the five variables that span it (mean/max/min tem-
 157 perature, humidity, dew point); solar radiation and the UV index, which
 158 lack a pre-2014 record, fall back to a Theil–Sen fit over their available 2014–
 159 2024 window. The ensembles are trained on the deseasonalised residual
 160 $r(t) = y(t) - \hat{\beta}t$, so they model only the seasonal signal; the linear trend
 161 is added back at prediction time, $\hat{y}(t) = \hat{r}(t) + \hat{\beta}t$. This design makes each
 162 projected year differ from the previous one by exactly $\hat{\beta}$ and guarantees that
 163 the projected inter-annual tendency equals the reported Mann–Kendall/Sen
 164 slope (Section 2.12). We emphasise that this trend term is included for
 165 physical consistency with the observed record, not to improve short-horizon
 166 accuracy: at monthly resolution the seasonal cycle dominates, so the trend
 167 contributes only a small secular adjustment.

168 The winning model per variable (Section 2.6) is refit on the full 2014–2024
 169 residual and projected forward for a 24-month horizon (January 2025 through

December 2026). Projected primary variables are clipped to physically plausible bounds before any downstream computation to prevent non-physical extrapolations from propagating into the indices.

2.6. Model evaluation and selection

Reported test metrics use a strictly chronological hold-out: models are trained on 2014–2022 and tested on the withheld 2023–2024 period. Generalisation is additionally assessed by five-fold `TimeSeriesSplit` cross-validation on the residual series. For each candidate model we record the coefficient of determination R^2 and root-mean-square error (RMSE) on the chronological test set, the cross-validated R^2 ($\text{CV-}R^2$), a generalisation gap $\text{Gap} = |R^2 - \text{CV-}R^2|$, and an operational tolerance accuracy A_τ defined as the fraction of test months whose absolute error does not exceed a variable-specific tolerance τ ,

$$A_\tau = \frac{1}{N} \sum_{n=1}^N \mathbf{1}(|y_n - \hat{y}_n| \leq \tau). \quad (3)$$

The winning model is chosen by Simple Additive Weighting (SAW), a multi-attribute decision-making scheme [11], with a generalisation-gap penalty. Each criterion c is min–max normalised across the four candidates—directly for benefit criteria and by complement for cost criteria,

$$\tilde{s}_c = \begin{cases} \frac{s_c - \min_c}{\max_c - \min_c}, & c \text{ maximised,} \\ \frac{\max_c - s_c}{\max_c - \min_c}, & c \text{ minimised,} \end{cases} \quad (4)$$

and the composite score is the weighted sum $S = \sum_c w_c \tilde{s}_c$, with weights $w_{R^2} = 0.25$, $w_{\text{CV-}R^2} = 0.25$, $w_{\text{RMSE}} = 0.20$, $w_{\text{Gap}} = 0.20$, and $w_{A_\tau} = 0.10$ (RMSE and Gap treated as cost criteria). The highest-scoring model wins; models with $\text{Gap} > 0.10$ are additionally flagged as overfitting risks.

191 2.7. Thermal index computation

192 To avoid target leakage, thermal-stress indices are never predicted di-
 193 rectly; they are recomputed analytically from the *projected* primary vari-
 194 ables. The heat index (HI) uses the U.S. National Weather Service Rothfusz
 195 regression [18]. With temperature T_F and relative humidity R (in °F and
 196 %),

$$\begin{aligned} \text{HI} = & -42.379 + 2.04901523 T_F + 10.14333127 R - 0.22475541 T_F R \\ & - 6.83783 \times 10^{-3} T_F^2 - 5.481717 \times 10^{-2} R^2 + 1.22874 \times 10^{-3} T_F^2 R \\ & + 8.5282 \times 10^{-4} T_F R^2 - 1.99 \times 10^{-6} T_F^2 R^2, \end{aligned} \quad (5)$$

197 with the standard low-humidity and high-humidity adjustments applied within
 198 their validity ranges and a simplified formula substituted below 80°F; the
 199 result is returned in °C. Wet-bulb temperature T_w follows the empirical
 200 formulation of (author?) [19],

$$\begin{aligned} T_w = & T \arctan[0.151977 (R + 8.313659)^{1/2}] + \arctan(T + R) \\ & - \arctan(R - 1.676331) + 0.00391838 R^{3/2} \arctan(0.023101 R) - 4.686035, \end{aligned} \quad (6)$$

201 with T in °C and R in %. Wet-bulb globe temperature is computed in a
 202 shade variant (no direct-beam term) and a sun variant that adds a solar-
 203 radiation contribution S ,

$$\text{WBGT}_{\text{shade}} = 0.7 T_w + 0.3 T, \quad (7)$$

$$\text{WBGT}_{\text{sun}} = 0.7 T_w + 0.2 (T + 0.025 S) + 0.1 T. \quad (8)$$

204 Cooling degree-days are expressed as a monthly-mean daily rate about an
 205 18°C base,

$$\text{CDD} = \max\left(\frac{T_{\text{max}} + T_{\text{min}}}{2} - 18, 0\right). \quad (9)$$

206 A composite heatstroke-hazard score (a sigmoid of the heat index mod-
 207 ulated by a wet-bulb multiplier) was also implemented and recalibrated for
 208 monthly-mean inputs. However, in external validation it produced a nega-
 209 tive out-of-sample R^2 (i.e. worse than a constant predictor), so it is *excluded*
 210 *from all predictive results* and retained only as a descriptive projected field.
 211 This exclusion is stated explicitly to avoid over-interpretation of an index
 212 that does not demonstrate skill.

213 2.8. Operational heat-stress conditions and risk categorisation

214 The projected fields are organised into seven temperature-correlated heat-
 215 stress conditions of operational relevance: (1) heat-index comfort (perceived
 216 temperature under humidity); (2) wet-bulb temperature (the thermody-
 217 namic limit of evaporative cooling); (3) wet-bulb globe temperature in in-
 218 door (shade) and outdoor (sun) forms (the occupational heat-exposure stan-
 219 dard); (4) heatstroke risk; (5) cooling degree-days (a building-cooling en-
 220 ergy proxy); (6) ultraviolet exposure risk; and (7) dew-point moisture stress.
 221 Conditions (6) and (7) are obtained directly from the projected UV-index
 222 and dew-point fields, which are among the primary modelled variables (Sec-
 223 tion 2.2); condition (4) is excluded from predictive results as noted above.

224 To move from continuous values to operational meaning, four of the
 225 conditions are additionally mapped to established risk-category systems:
 226 the U.S. National Weather Service heat-index comfort classes (Comfortable,
 227 Caution, Extreme Caution, Danger, Extreme Danger); WBGT occupational-
 228 exposure classes (Low, Moderate, High, Extreme); the World Health Orga-
 229 nization UV-index classes (Low, Moderate, High, Very High, Extreme) [20];
 230 and dew-point human-comfort classes (Dry through Extremely Oppressive).
 231 Wet-bulb temperature (2) and cooling degree-days (5) are reported as con-

tinuous physical quantities. The category boundaries are conventionally defined for daily peak values; because the present pipeline operates on monthly means, the extreme classes are seldom reached and the categories should be read as describing *typical monthly* conditions rather than daily peaks. Categorisation skill is assessed against the observed 2025 fields by exact-match agreement and by agreement to within one adjacent class.

2.9. Uncertainty quantification

Projection uncertainty for the temperature–humidity indices is obtained by first-order (linearised) error propagation from the primary-variable uncertainties. For an index $I(x_1, \dots, x_k)$ with input standard deviations σ_{x_j} ,

$$\sigma_I = \left[\sum_j \left(\frac{\partial I}{\partial x_j} \right)^2 \sigma_{x_j}^2 \right]^{1/2}, \quad (10)$$

where the input σ_{x_j} are taken from each winning model’s in-sample RMSE and representative sensitivity coefficients are used for the heat index, wet-bulb temperature, and WBGT. The propagated σ_I define the 68% ($\pm\sigma_I$) and 95% ($\pm 1.96\sigma_I$) bands reported for the projected indices.

2.10. Baseline skill benchmarks

Model skill is benchmarked against two references evaluated on the same chronological test set: a seasonal-naive predictor (each test month set to the same calendar month one year earlier) and a monthly-climatology predictor (the training-period mean for each calendar month). Skill relative to a reference is

$$SS = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{ref}}}, \quad (11)$$

so that positive values indicate improvement over the reference.

253 *2.11. External validation against observed 2025*

254 The 24-month projection includes calendar year 2025, for which inde-
 255 pendent observations were withheld from all training. Projected monthly
 256 primary variables and recomputed indices are compared with the observed
 257 2025 monthly series using R^2 , RMSE, mean absolute error, and the tolerance-
 258 accuracy bands of Section 2.6. The final month (November 2025) is partial
 259 in the source data and is treated accordingly. This constitutes a genuine,
 260 fully out-of-sample test of the projection.

261 *2.12. Observational trend analysis*

262 Long-term trends are characterised on the full 1980–2024 annual-mean
 263 series using the non-parametric Mann–Kendall test with Theil–Sen slope
 264 estimation, computed independently of the predictive models (and therefore
 265 free of target leakage). The Mann–Kendall statistic is

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(x_j - x_k), \quad (12)$$

266 with tie-corrected variance

$$\text{Var}(S) = \frac{n(n-1)(2n+5) - \sum_i t_i(t_i-1)(2t_i+5)}{18}, \quad (13)$$

267 where t_i is the number of ties of extent i . The standardised statistic

$$Z = \begin{cases} (S-1)/\sqrt{\text{Var}(S)}, & S > 0, \\ 0, & S = 0, \\ (S+1)/\sqrt{\text{Var}(S)}, & S < 0, \end{cases} \quad (14)$$

268 is referred to the standard normal distribution for a two-sided p -value, and
 269 Kendall’s $\tau = 2S/[n(n-1)]$ measures rank correlation with time. The

270 Theil–Sen slope (Eq. 2) provides the trend magnitude, reported per decade.
 271 Trends with $p < 0.05$ are deemed significant. This analysis is performed
 272 for mean, maximum, and minimum temperature, humidity, dew point, and
 273 the temperature–humidity indices (heat index, wet-bulb temperature, shade
 274 WBGT) that are computable across the full record.

275 A subset of districts encode pre-~2000 missing days as zero in the source
 276 record; left uncorrected, the resulting $0 \rightarrow \sim 30^\circ\text{C}$ step fabricates physically
 277 impossible trends (an artefactual $+7.8^\circ\text{C}$ per decade in one district). Before
 278 any trend is fit, daily values below physically plausible floors (5°C for tem-
 279 peratures, with lower floors for minimum temperature and dew point, and
 280 5% for humidity) are therefore treated as missing, and a year must retain at
 281 least 60 valid days to enter the annual-mean series. The same masked slopes
 282 are used when the trend is re-added to the projection (Section 2.5). This
 283 affects only the observational trend and long-run slope; the machine-learning
 284 models are unaffected because they train exclusively on the complete 2014–
 285 2024 window.

286 2.13. Zonal synthesis

287 District-level projections are summarised by zone as monthly and annual
 288 envelopes (the minimum, mean, and maximum across member districts) for
 289 the key indices, providing compact, journal-page-efficient views of the pro-
 290 jected heat-stress distribution in place of 64 individual district panels. Full
 291 district-level detail is additionally retained through choropleth maps ren-
 292 dered directly from the CC BY 3.0 IGO geoBoundaries Bangladesh ADM2
 293 administrative boundaries [21]; districts are matched to the boundary geom-
 294 etry by name, and per-district values are mapped for the observed trends, the
 295 projected annual means of all seven conditions, and the number of months

per year each district spends in each condition’s hazard class.

2.14. Implementation and reproducibility

The pipeline is implemented in Python 3.14 using scikit-learn 1.8 [22], LightGBM 4.6, XGBoost 3.3, CatBoost 1.2.10, together with NumPy, pandas, and Matplotlib. All stochastic components use a fixed seed, and a run manifest records package versions, configuration constants, and per-run timings so that every reported artefact is traceable to a specific configuration.

3. Results

The full pipeline was executed for all 64 districts, producing 448 district-variable models, a 24-month projection (January 2025–December 2026) per district, an out-of-sample validation against observed 2025, and the 1980–2024 observational trend analysis. Unless stated otherwise, statistics are reported as the median across the 64 districts.

3.1. Predictive accuracy and model selection

On the chronological 2023–2024 hold-out, the ensembles reproduced the primary variables accurately, with median test R^2 of 0.970 for minimum temperature, 0.961 for mean temperature, 0.962 for dew point, and 0.886 for maximum temperature; humidity (0.632), solar radiation (0.691), and the UV index (0.664) were intrinsically harder and scored lower. Under the SAW criterion, LightGBM was selected most frequently (284 of 448 fits), followed by Random Forest (111) and CatBoost (53); XGBoost, although included in the candidate pool, was never top-ranked. Model preference was variable-dependent (Table 1): LightGBM dominated temperature (57/64), maximum temperature (63/64), and dew point (48/64), whereas Random Forest was

320 preferred for the noisier humidity field (55/64) and was competitive for solar
 321 radiation and UV.

Table 1: Best-model selection frequency across the 64 districts, by variable (SAW criterion). XGBoost was never selected and is omitted.

Variable	LightGBM	Random Forest	CatBoost
Mean temperature	57	0	7
Maximum temperature	63	0	1
Minimum temperature	44	2	18
Relative humidity	4	55	5
Dew point	48	0	16
Solar radiation	29	29	6
UV index	39	25	0

322 3.2. External validation against observed 2025

323 The projected 2025 fields, produced without any exposure to 2025 ob-
 324 servations, validated strongly (Table 2). Median R^2 reached 0.982 for min-
 325 imum temperature, 0.980 for dew point, and 0.950 for mean temperature
 326 (RMSE 0.72 °C). Crucially, the thermal-stress indices—recomputed from
 327 the projected primary variables rather than predicted directly—also vali-
 328 dated well: wet-bulb temperature and shade WBGT both at $R^2 = 0.969$
 329 (RMSE ≤ 0.63 °C), sun WBGT at 0.966, the heat index at 0.951 (RMSE
 330 1.36 °C), and cooling degree-days at 0.951. Maximum temperature (0.845),
 331 humidity (0.825), UV (0.805), and solar radiation (0.766) were the weakest,
 332 consistent with their lower learnability. For the six exemplar districts, heat-
 333 index validation R^2 ranged from 0.873 (Sylhet) and 0.891 (Chittagong) to

334 0.948–0.954 (Kushtia, Khagrachhari, Rajshahi, Dhaka), with RMSE between
335 1.1 and 1.9 °C.

Table 2: External validation of the projected 2025 fields against observed 2025, summarised as the median across 64 districts. Errors are in each variable’s native unit (°C for temperatures and thermal indices; % for humidity; W m^{−2} for solar radiation; index units for UV; °C d d^{−1} for CDD).

Variable	n	Median R^2	Median RMSE	Median MAE
Mean temperature	64	0.950	0.720	0.615
Maximum temperature	64	0.845	0.940	0.788
Minimum temperature	64	0.982	0.565	0.449
Relative humidity	64	0.825	2.334	1.996
Dew point	64	0.980	0.601	0.469
Solar radiation	64	0.766	14.352	11.704
UV index	64	0.805	0.375	0.302
Heat index	64	0.951	1.357	1.142
Wet-bulb temperature	64	0.969	0.624	0.509
WBGT (shade)	64	0.969	0.595	0.506
WBGT (sun)	64	0.966	0.644	0.543
Cooling degree-days	64	0.951	0.696	0.577

336 3.3. Skill relative to baselines

337 Against the seasonal-naive reference the ensembles were skilful in 81.5%
338 of district–variable cases, with the largest median gains for minimum tem-
339 perature (+0.24) and dew point (+0.21). Relative to monthly climatology,
340 however, the models were essentially at parity: the median skill score was

341 near zero or slightly negative for every variable (e.g. -0.003 for mean tem-
 342 perature, -0.030 for maximum temperature, -0.059 for minimum temper-
 343 ature), and only 31.9% of cases exceeded climatology. This is the expected
 344 behaviour at monthly resolution: the ensembles reconstruct the strong trop-
 345 ical seasonal cycle that monthly climatology already encodes, so they add
 346 little in raw error terms. Their contribution is therefore not a reduction in
 347 monthly RMSE but a spatially complete, uncertainty-quantified, and trend-
 348 consistent projection that climatology cannot provide.

349 *3.4. Observed long-term trends, 1980–2024*

350 The Mann–Kendall/Sen analysis revealed a coherent warming signal (Ta-
 351 ble 3, Figure 2). Maximum temperature increased significantly in all 64 dis-
 352 tricts (median $+0.56^\circ\text{C}$ per decade), and the heat index, wet-bulb temper-
 353 ature, and shade WBGT increased in 58, 57, and 58 of 64 districts (median
 354 $+0.37$, $+0.17$, and $+0.17^\circ\text{C}$ per decade, respectively). Mean temperature
 355 rose in 50 districts and dew point in 46. Minimum temperature was markedly
 356 more heterogeneous (16 increasing, 13 decreasing, 35 without significant
 357 trend; median $+0.02^\circ\text{C}$ per decade), and humidity was mixed (24 increas-
 358 ing, 23 decreasing). The sharp contrast between near-universal maximum-
 359 temperature warming and the essentially flat minimum-temperature signal
 360 indicates a widening diurnal temperature range in much of the country; a
 361 small number of districts (e.g. Sylhet) therefore show a slight decrease in
 362 *mean* temperature despite rising maxima. Because the projection re-imposes
 363 exactly these Sen slopes, the projected inter-annual tendency reproduces this
 364 observed signal by construction.

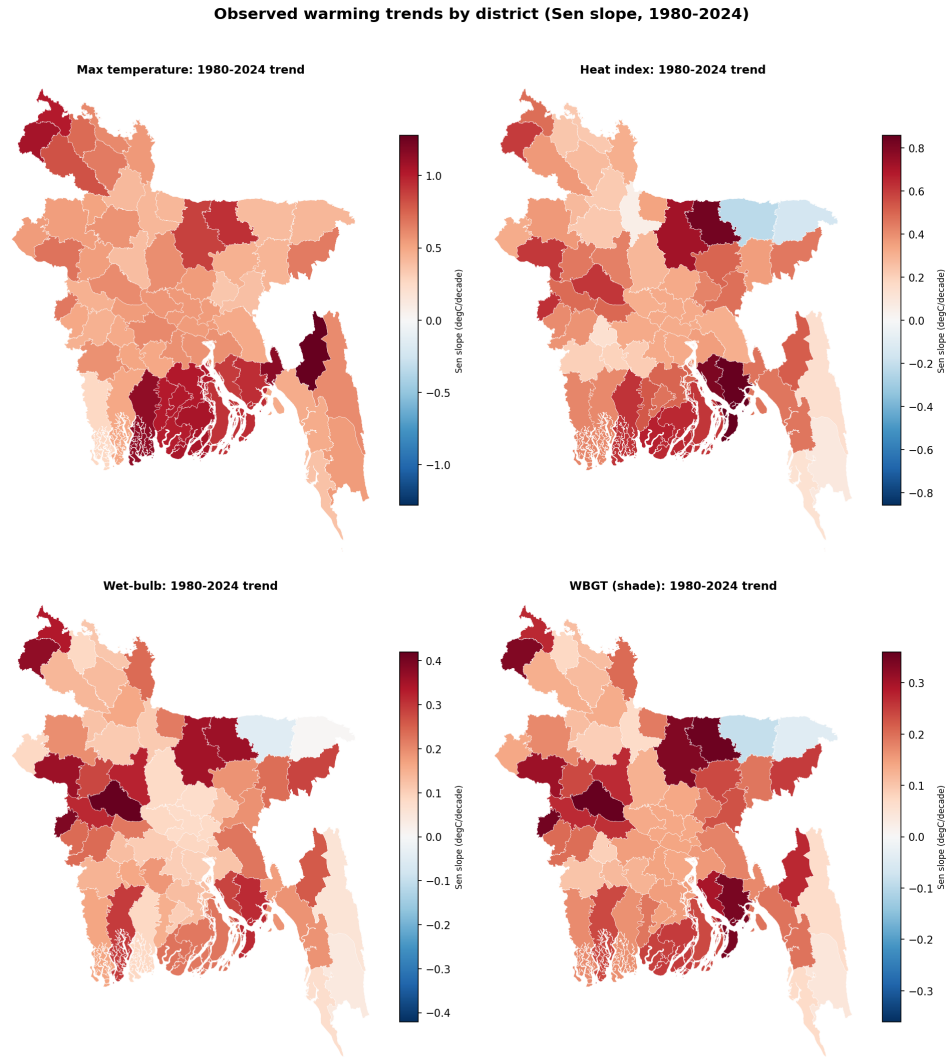


Figure 2: Observed 1980–2024 warming by district (Theil-Sen slope, °C per decade) for maximum temperature, heat index, wet-bulb temperature, and shade WBGT. Red indicates warming, blue cooling. Values are computed after masking missing-as-zero early-record days (Section 2.12).

Table 3: Observed 1980–2024 trends (Mann–Kendall test with Theil–Sen slope) by variable: number of districts increasing, decreasing, or without a significant trend ($p < 0.05$), and the median Sen slope per decade ($^{\circ}\text{C}$ per decade, except humidity in % per decade).

Variable	Increasing	Decreasing	No trend	Median slope/decade
Maximum temperature	64	0	0	+0.558
Mean temperature	50	2	12	+0.194
Minimum temperature	16	13	35	+0.021
Dew point	46	0	18	+0.197
Relative humidity	24	23	17	+0.177
Heat index	58	1	5	+0.373
Wet-bulb temperature	57	0	7	+0.165
WBGT (shade)	58	0	6	+0.172

3.5. Projected 2025–2026 outlook and zonal synthesis

Consistent with the modest observed slopes, the projected year-over-year change was small: the heat index rose by a median of $+0.038^{\circ}\text{C}$ per year across the 64 districts (range -0.034 to $+0.096^{\circ}\text{C}$ per year), the negative values corresponding to the districts whose 1980–2024 mean-temperature trend is itself slightly negative. The zonal synthesis (Table 4) places the highest projected annual-mean heat index in the Southwest Hot–Dry Zone (31.0°C) and the Coastal Humid Heat Zone (30.6°C), and the lowest in the Haor/Wetland Zone (28.0°C); sun WBGT peaked in the coastal belt (25.5°C). The two continuous-burden conditions reinforce this ordering: projected cooling degree-days—a proxy for building cooling-energy demand—were highest in the Southwest Hot–Dry and Coastal Humid Heat zones (9.1 and $9.0^{\circ}\text{Cdd}^{-1}$) and lowest in the Haor/Wetland Zone (7.0), while

378 projected wet-bulb temperature, the thermodynamic ceiling on evaporative
 379 cooling, followed the same spatial pattern, peaking in the Coastal (23.7°C)
 380 and Southwest (23.6°C) zones and again reaching its minimum in the Haor
 381 basin (22.6°C). The coastal belt therefore emerges as the zone of great-
 382 est combined perceived-heat, humid-heat, and cooling-energy burden. Fig-
 383 ure 6 maps all six continuous conditions at district resolution, confirming the
 384 coastal-south concentration of heat, humidity, and cooling demand. Figure 3
 385 illustrates the full record-validation-projection narrative for the Dhaka ex-
 386 emplar, and Figure 4 summarises the projected monthly climatology of all
 387 six key conditions across the six zones, exposing their distinct seasonal sig-
 388 natures. Figure 5 shows the district spread within the hottest (Southwest)
 389 zone.

Table 4: Projected annual-mean thermal indices by climatic zone (zone mean, with
 the across-district minimum and maximum in parentheses). Values in °C except CDD
 (°Cdd⁻¹).

Zone	Heat index	WBGT (sun)	Wet-bulb	CDD
1. Northwest Extreme Heat	29.6 (28.4–30.7)	24.7	22.9	8.3
2. Central Urban Heat	29.6 (29.3–30.9)	24.8	22.8	8.8
3. Southwest Hot–Dry	31.0 (30.6–31.9)	25.5	23.6	9.1
4. Coastal Humid Heat	30.6 (29.5–31.5)	25.5	23.7	9.0
5. Haor/Wetland Humid	28.0 (26.0–29.5)	24.3	22.6	7.0
6. Hill Tract	29.9 (29.3–30.3)	25.4	23.5	8.6

390 3.6. Operational risk categories

391 Mapping the projected fields onto their risk-category systems yields a
 392 policy-facing summary of the seven conditions (Table 5). Under monthly-

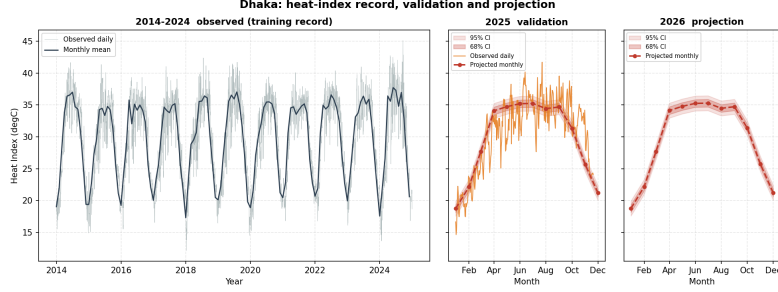


Figure 3: Heat-index record, validation, and projection for the Dhaka exemplar: (left) observed daily heat index 2014–2024 with the monthly mean; (centre) the 2025 validation year, observed daily versus the monthly projection with 68% and 95% confidence bands; (right) the 2026 monthly projection with confidence cloud.

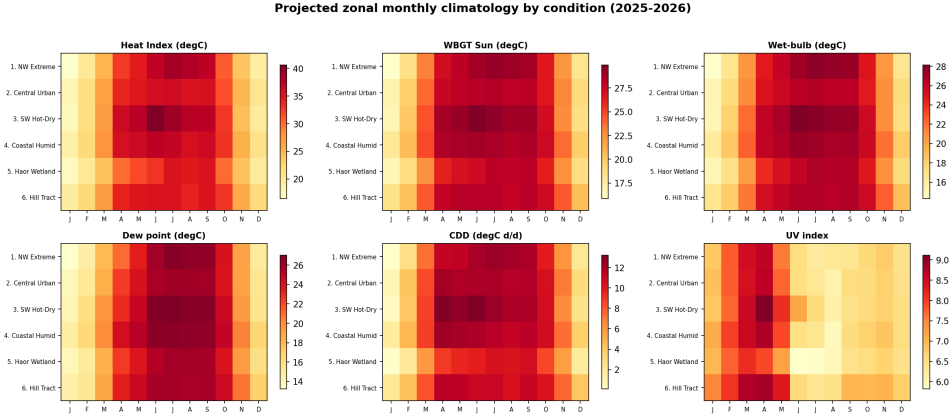


Figure 4: Projected zonal monthly climatology for six heat-stress conditions (heat index, sun WBGT, wet-bulb temperature, dew point, cooling degree-days, and UV index). The temperature-driven conditions peak in the pre-monsoon to monsoon window, whereas the UV index peaks in the dry pre-monsoon (March–April) and falls during the cloudy monsoon — a distinct, radiation-driven seasonality.

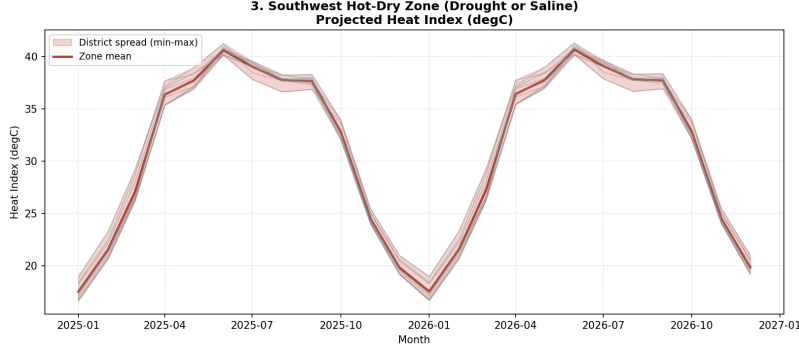


Figure 5: Projected heat index for the Southwest Hot-Dry Zone: the zone mean and the min-max envelope across member districts, illustrating within-zone spread.

mean heat index, projected months split between Comfortable (37.8%) and Extreme Caution (51.8%), with the intermediate Caution class at 10.2% and the Danger class essentially absent (0.1%)—the latter a direct consequence of categorising monthly means rather than daily peaks. Outdoor WBGT was predominantly Low (59.0%) to Moderate (37.0%), reaching the High class in 3.9% of months, while indoor (shade) WBGT rarely left the Low class (79.8%). The UV signal was the most consistently hazardous condition: 75.5% of projected months fall in the WHO High class and a further 20.4% in Very High. Dew-point moisture stress was also severe, with 45.5% of months in the Extremely Oppressive class. Across zones, the Southwest Hot-Dry Zone carried the largest Extreme-Caution share for heat index (57.5%) and the only appearance of the Danger class (0.8%), whereas the Haor/Wetland Zone was the mildest (35.4% Extreme Caution); Figure 7 shows the full zonal composition and Figure 8 maps, for each district, the number of months per year spent at or above each condition’s hazard threshold. The months-per-year view resolves the strong spatial and seasonal gradients that a dominant-class map obscures: the coastal south en-

Projected 2025-2026 annual-mean heat-stress conditions by district

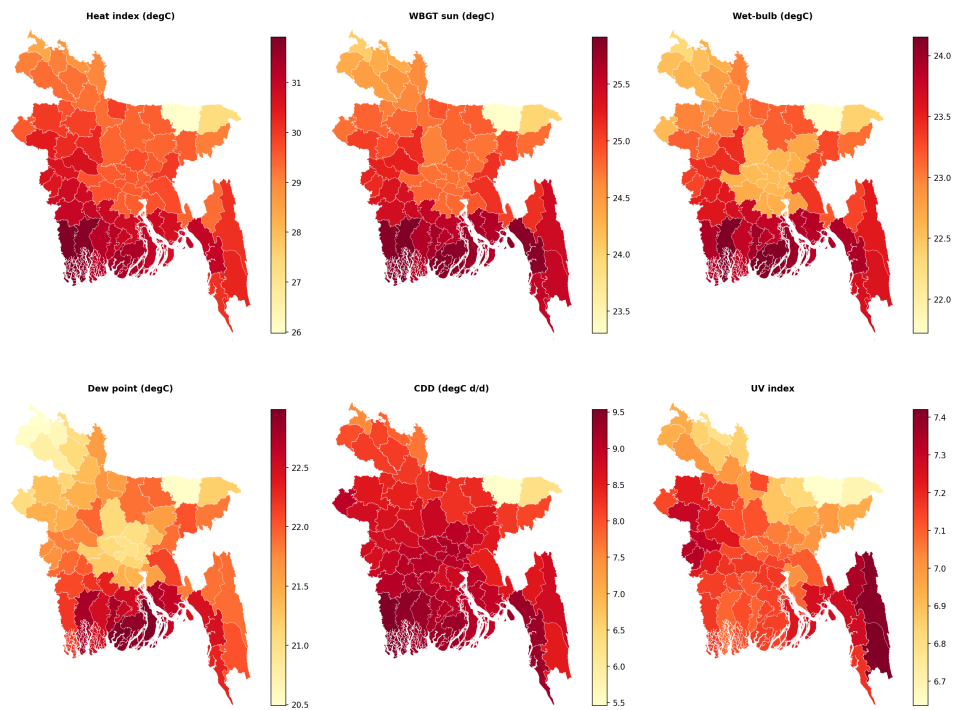


Figure 6: Projected 2025–2026 annual-mean heat-stress conditions by district: heat index, sun WBGT, wet-bulb temperature, dew point, cooling degree-days, and UV index. Every condition is shown at full district resolution rather than aggregated to zones.

dures dew-point levels in the Extremely Oppressive class for roughly seven to eight months per year against three to four in the northwest, and very-high UV persists longest in the south.

Critically, the categorical layer validated well against observed 2025 (Table 6): exact-class agreement ranged from 81.5% (UV) to 89.8% (heat-index comfort), and agreement to within one adjacent class reached 100% for all five categorised conditions. The projection therefore reproduces not only the continuous fields but also their operational risk classification.

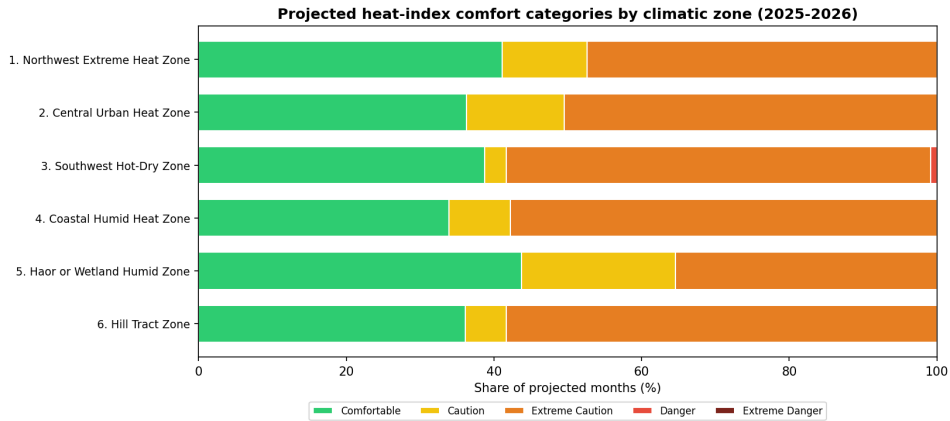


Figure 7: Projected heat-index comfort categories by climatic zone (share of projected months, 2025–2026). The Southwest Hot–Dry Zone carries the largest Extreme-Caution share and the only Danger occurrences; the Haor/Wetland Zone is mildest.

3.7. Projection uncertainty

The first-order error propagation yielded confidence bands that are approximately constant across the 24-month horizon, reflecting the dominance of the seasonal model’s error over the small trend extrapolation. For the Dhaka exemplar, the 95% heat-index band was $\pm 1.17^\circ\text{C}$, the wet-bulb band $\pm 0.99^\circ\text{C}$, and the sun-WBGT band $\pm 0.76^\circ\text{C}$. These bands exceed the pro-

Table 5: Projected risk-category distribution (share of projected district-months, 2025–2026, 64 districts). Only populated categories are shown.

Condition	Category	Share (%)
Heat-index comfort	Comfortable	37.8
	Caution	10.2
	Extreme Caution	51.8
	Danger	0.1
WBGT (sun, outdoor)	Low	59.0
	Moderate	37.0
	High	3.9
WBGT (shade, indoor)	Low	79.8
	Moderate	20.2
UV (WHO)	Moderate	4.2
	High	75.5
	Very High	20.4
Dew-point moisture stress	Comfortable	19.7
	Slightly Humid	6.6
	Humid	16.3
	Very Humid / Oppressive	11.9
	Extremely Oppressive	45.5

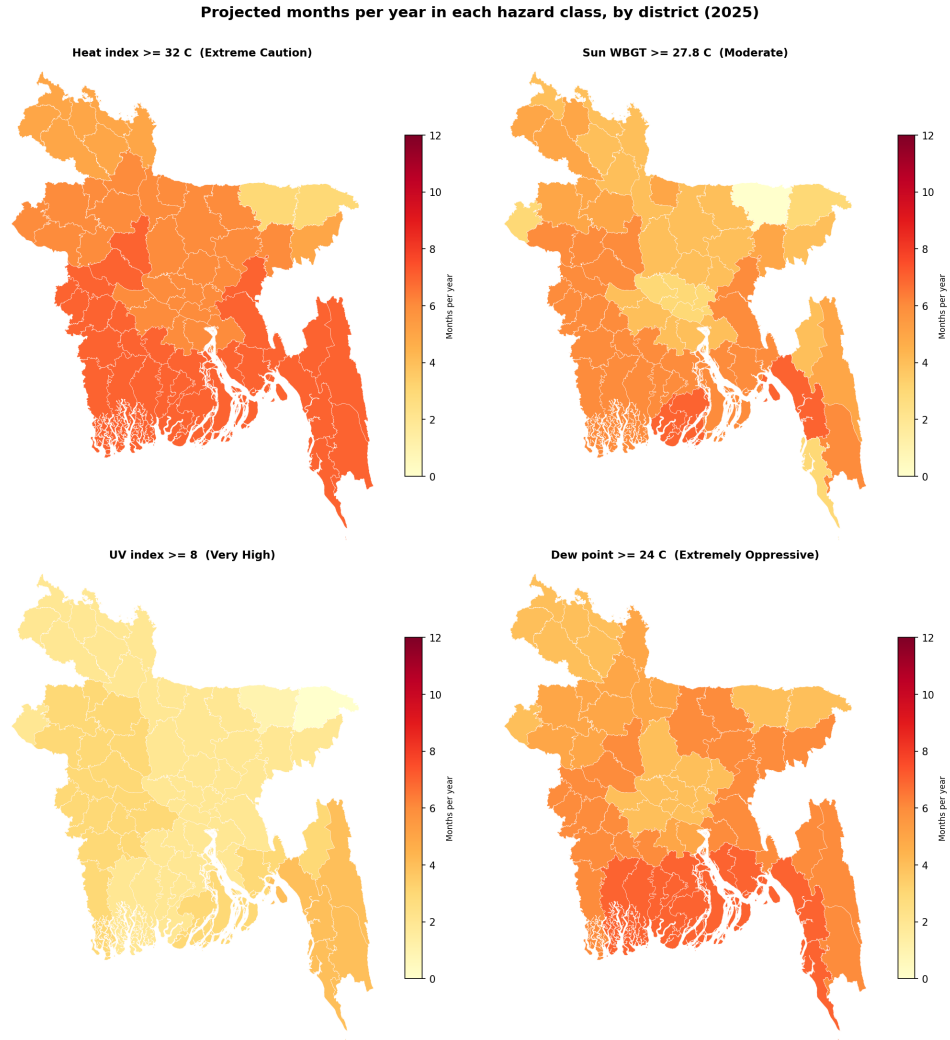


Figure 8: Projected number of months per year (of 12) that each district spends at or above a hazard threshold, for heat index ($\geq 32^{\circ}\text{C}$, Extreme Caution), sun WBGT ($\geq 27.8^{\circ}\text{C}$, Moderate), UV (≥ 8 , Very High), and dew point ($\geq 24^{\circ}\text{C}$, Extremely Oppressive). This months-per-year measure preserves the seasonal and spatial structure that a single dominant-class map collapses.

Table 6: Categorical validation against observed 2025: exact-class agreement and agreement to within one adjacent class, pooled over district-months.

Condition	n	Exact-class (%)	Within one class (%)
Heat-index comfort	704	89.8	100.0
WBGT (sun, outdoor)	704	84.1	100.0
WBGT (shade, indoor)	704	88.8	100.0
UV (WHO)	704	81.5	100.0
Dew-point moisture stress	704	87.1	100.0

jected inter-annual trend signal by more than an order of magnitude, and this relationship is stated explicitly to caution against over-interpreting small year-to-year differences in the projection.

4. Discussion

4.1. Principal findings

This study delivers district-resolved, uncertainty-aware near-term projections of seven temperature-correlated heat-stress conditions for all 64 districts of Bangladesh, together with a leakage-free external validation against independent 2025 observations. The recomputed thermal indices validated strongly—heat index, wet-bulb temperature, and WBGT all at median $R^2 \geq 0.95$ —and, when mapped onto their operational risk classes, reproduced the observed 2025 class in 81–90% of district-months exactly and in 100% to within one adjacent class. Because the indices are computed from the projected primary variables rather than predicted directly, this accuracy is achieved without target leakage and while preserving the physical coupling among temperature, humidity, and radiation. **[PLACEHOLDER: posi-**

440 tioning relative to prior heat-projection studies for Bangladesh /
441 South Asia.]

442 *4.2. Skill must be interpreted correctly*

443 A central and deliberately transparent result is that, although the ensem-
444 bles comfortably outperform a seasonal-naive baseline, they are essentially
445 at parity with monthly climatology (positive skill in only $\sim 32\%$ of district-
446 variable cases). This is expected rather than disappointing: at monthly
447 resolution the tropical seasonal cycle dominates the variance, and any com-
448 petent learner reconstructs the climatological annual march that climatology
449 also encodes. The contribution of the framework is therefore not a reduc-
450 tion in monthly error but the provision of a spatially complete, uncertainty-
451 quantified, and trend-consistent projection—including physiologically rele-
452 vant indices and their operational risk classes—that a bare climatology can-
453 not supply. We state this explicitly to discourage over-claiming predictive
454 skill that the data do not support. **[PLACEHOLDER: contrast with**
455 **studies that report ML skill without a climatology baseline.]**

456 *4.3. Physical and spatial interpretation*

457 The observational analysis reveals a coherent but spatially structured
458 warming signal. Maximum temperature increased in every district (median
459 $+0.56^\circ\text{C}$ per decade) while minimum temperature remained essentially flat
460 (median $+0.02^\circ\text{C}$ per decade), implying a widening diurnal temperature
461 range across much of the country—a pattern with distinct implications for
462 daytime heat exposure and nighttime physiological recovery. Humid-heat
463 indices (heat index, wet-bulb, WBGT) rose in 57–58 of 64 districts, con-
464 firming that the warming is not purely a dry-bulb phenomenon. Spatially,

the Southwest Hot–Dry and Coastal Humid Heat zones carry the greatest projected burden across perceived heat, humid heat, and cooling-energy demand, whereas the elevated Haor/Wetland basin is consistently mildest; the coastal belt in particular combines the highest wet-bulb temperatures and cooling degree-days, marking it as a priority region. **[PLACEHOLDER: link the diurnal-range and coastal-humidity findings to physiological and epidemiological literature on wet-bulb thresholds.]**

4.4. Operational risk categories and their message

Translating the projections into established risk classes sharpens the public-health narrative. The UV signal is the most uniformly hazardous condition: essentially the entire country falls in the WHO High class or above throughout the projection, and dew-point moisture stress reaches the Extremely Oppressive class in almost half of all projected months. Heat-index comfort is dominated by the Extreme Caution class in the warm season, while the Danger class is essentially absent—a direct and important consequence of categorising monthly means rather than daily maxima (Section 2.8), which the categories must be read against. The nationwide reach of the UV and humidity hazards, contrasted with the more spatially variable heat-index and WBGT burden, argues for differentiated adaptation responses. **[PLACEHOLDER: connect UV and oppressive-humidity exposure to health outcomes and occupational-safety guidance.]**

4.5. Methodological reflections

Three design choices proved consequential. First, recomputing indices from projected primary variables (rather than modelling them directly) both prevents target leakage and guarantees internal physical consistency of the

490 projected fields. Second, the detrend–residual–retrend construction resolves
 491 a failure mode of tree ensembles—their inability to extrapolate a calendar-
 492 year feature—so that projected years evolve with the observed climate trend
 493 instead of collapsing onto the last training year; imposing the robust 1980–
 494 2024 Sen slope additionally keeps the projection consistent with the manuscript’s
 495 own trend analysis by construction. Third, the district choropleths ex-
 496 posed a data-quality problem (early-record missing values encoded as zero)
 497 that zonal-median summaries had masked, underscoring the value of full-
 498 resolution spatial diagnostics and prompting the physical-floor correction
 499 described in Section 2.12.

500 *4.6. Limitations*

501 Several limitations bound the interpretation. (i) The projection horizon
 502 is deliberately restricted to 24 months; because the re-added trend is small
 503 relative to the projection uncertainty (the 95% heat-index band exceeds the
 504 annual trend signal by more than an order of magnitude), longer horizons
 505 would add deterministic drift without validatable information. (ii) Work-
 506 ing at monthly-mean resolution necessarily compresses daily extremes, so
 507 the risk categories describe typical monthly conditions and systematically
 508 under-represent short-duration dangerous events; a daily-resolution exten-
 509 sion is the natural next step. (iii) A composite heatstroke-hazard index was
 510 implemented but excluded from predictive results for negative out-of-sample
 511 skill, and is reported only descriptively. (iv) All meteorological inputs de-
 512 rive from a single commercial reanalysis-blended source; despite strong 2025
 513 validation, independent station-based corroboration would strengthen confi-
 514 dence, and the early-record inhomogeneity in a subset of districts (mitigated
 515 here by masking) illustrates the sensitivity of long-run trends to input qual-

ity. (v) The models add little skill over climatology at monthly scale, as discussed above. **[PLACEHOLDER: compare these limitations and the horizon choice with dynamical-downscaling and CMIP-based projection studies.]**

4.7. Outlook

[PLACEHOLDER: future-work paragraph: daily-resolution extension, integration with health and energy-demand data, and comparison with dynamical projections — to be completed with the literature review.]

5. Conclusion

This study introduced the Temperature-Related Conditions (TRC) framework and applied it to project seven physiologically and operationally relevant heat-stress conditions across all 64 districts of Bangladesh. By training gradient-boosting ensembles on the 2014–2024 record with future-known seasonal features, re-imposing the observed 1980–2024 trend, and recomputing every thermal index from the projected primary variables, the framework produced district-level 24-month projections that validated strongly against independent 2025 observations—median R^2 of 0.95–0.98 for the key indices, with operational risk classes reproduced to within one category in 100% of district-months.

The observational analysis revealed a coherent but spatially structured warming signal: maximum temperature rose in every district (median $+0.56^\circ\text{C}$ per decade) while minimum temperature remained essentially flat, implying a widening diurnal temperature range, and the humid-heat indices increased across most of the country. Spatially, the coastal and southwestern zones

541 emerged as the areas of greatest combined perceived-heat, humid-heat, and
542 cooling-energy burden, whereas ultraviolet and oppressive-humidity hazards
543 were effectively nationwide. Crucially, the ensembles were found to be at
544 parity with monthly climatology; the framework’s contribution is therefore
545 not improved monthly accuracy but a spatially complete, uncertainty-aware,
546 and trend-consistent outlook—delivered at full district resolution—that sup-
547 ports targeted heat-adaptation planning. One indicator, heatstroke risk, was
548 excluded from the predictive results for a lack of out-of-sample skill and re-
549 ported only descriptively.

550 Several directions follow naturally. Extending the analysis to daily resolu-
551 tion would capture the short-duration extremes that monthly means smooth
552 away and would allow the risk categories to reflect daily peaks rather than
553 typical monthly conditions. Coupling the projected indices with health-
554 surveillance and energy-demand records would translate this physical outlook
555 into quantified public-health and infrastructure burdens **[verify-intended]**.
556 More broadly, the leakage-free, trend-consistent, and externally validated de-
557 sign demonstrated here offers a transferable template for district-level heat-
558 stress projection in other data-constrained, climatically heterogeneous re-
559 gions.

560 **CRedit authorship contribution statement**

561 **[PLACEHOLDER: CRedit roles per author.]**

562 **Declaration of competing interest**

563 **[PLACEHOLDER: competing-interest statement.]**

564 Data availability

565 Daily meteorological data are available from the Visual Crossing Weather
566 API (<https://www.visualcrossing.com>). **[PLACEHOLDER: final data-**
567 **availability wording and any access/licence details.]**

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